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Praktikum 2

Tugas mencari metode pencarian

1. buatlah pseudocode dan koding untuk algoritma red-black tree sesuai dengan ketentuan rule-nya.
2. analisa worst case, average case

Link kode Python saya: https://colab.research.google.com/drive/1YsMED-OCydGfP6lrCctCaexG-C_JiXpH?usp=sharing

The image displays two screenshots of a Google Colab notebook titled "G.231.24.0001_Rikabaguswijaya_Prak2_AKA.ipynb". The notebook is open in a web browser, showing the code editor and the output area.

Top Screenshot: The code editor shows the definition of a Node and the LEFT ROTATE function.

```
STRUKTUR NODE
Node:
data
color (RED / BLACK)
left
right
parent

LEFT ROTATE
LEFT-ROTATE(T, x):
    y = x.right
    x.right = y.left
    if y.left != NULL:
        y.left.parent = x
    y.parent = x.parent
    if x.parent == NULL:
```

Bottom Screenshot: The code editor shows the continuation of the LEFT ROTATE function and the definition of the RIGHT ROTATE function.

```

    y.parent = x.parent
    if x.parent == NULL:
        T.root = y
    else if x == x.parent.left:
        x.parent.left = y
    else:
        x.parent.right = y
    y.left = x
    x.parent = y

RIGHT ROTATE
RIGHT-ROTATE(T, y):
    x = y.left
    y.left = x.right
    if x.right != NULL:
        x.right.parent = y
    x.parent = y.parent
```

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```
[ ]
1 x.right.parent = y
2
3 x.parent = y.parent
4
5 if y.parent == NULL:
6     T.root = x
7 else if y == y.parent.right:
8     y.parent.right = x
9 else:
10    y.parent.left = x
11
12 x.right = y
13 y.parent = x
```

INSERT

```
[ ]
1 INSERT(T, z):
2     y = NULL
3     x = T.root
4
5     while x != NULL:
6         y = x
7         if z.data < x.data:
```

Variables Terminal

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```
[ ]
1 while x != NULL:
2     y = x
3     if z.data < x.data:
4         x = x.left
5     else:
6         x = x.right
7
8 z.parent = y
9
10 if y == NULL:
11     T.root = z
12 else if z.data < y.data:
13     y.left = z
14 else:
15     y.right = z
16
17 z.left = NULL
18 z.right = NULL
19 z.color = RED
20
21 INSERT-FIXUP(T, z)
```

INSERT FIXUP

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```
[ ] INSERT-FIXUP(T, z):
    while z.parent.color == RED:

        if z.parent == z.parent.parent.left:
            y = z.parent.parent.right

            if y.color == RED:
                z.parent.color = BLACK
                y.color = BLACK
                z.parent.parent.color = RED
                z = z.parent.parent

            else:
                if z == z.parent.right:
                    z = z.parent
                    LEFT-ROTATE(T, z)

                z.parent.color = BLACK
                z.parent.parent.color = RED
                RIGHT-ROTATE(T, z.parent.parent)

        else:
            (mirror dari kasus di atas)

    T.root.color = BLACK
```

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```
[ ] import random
import math

class Node:
    def __init__(self, data):
        self.data = data
        self.color = "RED"
        self.left = None
        self.right = None
        self.parent = None

class RedBlackTree:
    def __init__(self):
        self.TNULL = Node(0)
        self.TNULL.color = "BLACK"
        self.root = self.TNULL

    def left_rotate(self, x):
        y = x.right
        x.right = y.left
        if y.left != self.TNULL:
            y.left.parent = x
```

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```
[ ]
def right_rotate(self, y):
    if y.left != self.TNULL:
        y.left.parent = x

    y.parent = x.parent

    if x.parent == None:
        self.root = y
    elif x == x.parent.left:
        x.parent.left = y
    else:
        x.parent.right = y

    y.left = x
    x.parent = y

def right_rotate(self, y):
    x = y.left
    y.left = x.right

    if x.right != self.TNULL:
        x.right.parent = y

    x.parent = y.parent
```

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```
[ ]
x.right.parent = y

x.parent = y.parent

if y.parent == None:
    self.root = x
elif y == y.parent.right:
    y.parent.right = x
else:
    y.parent.left = x

x.right = y
y.parent = x

def insert(self, key):
    node = Node(key)
    node.left = self.TNULL
    node.right = self.TNULL

    y = None
    x = self.root

    while x != self.TNULL:
        y = x
        if node.data < x.data:
```

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```
while x != self.TNULL:
    y = x
    if node.data < x.data:
        x = x.left
    else:
        x = x.right

    node.parent = y

if y == None:
    self.root = node
elif node.data < y.data:
    y.left = node
else:
    y.right = node

node.color = "RED"
self.fix_insert(node)

def fix_insert(self, k):
    while k.parent and k.parent.color == "RED":
        if k.parent == k.parent.parent.right:
            u = k.parent.parent.left
            if u.color == "RED":
                u.color = "BLACK"
                k.parent.color = "BLACK"
                k.parent.parent.color = "RED"
                k = k.parent.parent
            else:
                if k == k.parent.left:
                    k = k.parent
                    self.right_rotate(k)
                k.parent.color = "BLACK"
                k.parent.parent.color = "RED"
                self.left_rotate(k.parent.parent)
        else:
            u = k.parent.parent.right
            if u.color == "RED":
                u.color = "BLACK"
                k.parent.color = "BLACK"
                k.parent.parent.color = "RED"
                k = k.parent.parent
            else:
                if k == k.parent.right:
                    k = k.parent
```

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```
while k.parent and k.parent.color == "RED":
    if k.parent == k.parent.parent.right:
        u = k.parent.parent.left
        if u.color == "RED":
            u.color = "BLACK"
            k.parent.color = "BLACK"
            k.parent.parent.color = "RED"
            k = k.parent.parent
        else:
            if k == k.parent.left:
                k = k.parent
                self.right_rotate(k)
            k.parent.color = "BLACK"
            k.parent.parent.color = "RED"
            self.left_rotate(k.parent.parent)
    else:
        u = k.parent.parent.right
        if u.color == "RED":
            u.color = "BLACK"
            k.parent.color = "BLACK"
            k.parent.parent.color = "RED"
            k = k.parent.parent
        else:
            if k == k.parent.right:
                k = k.parent
```

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```
k = k.parent.parent
else:
    if k == k.parent.right:
        k = k.parent
        self.left_rotate(k)
        k.parent.color = "BLACK"
        k.parent.parent.color = "RED"
        self.right_rotate(k.parent.parent)

    self.root.color = "BLACK"

# ===== FUNGSI ANALISIS =====
def get_height(self, node):
    if node == self.NULL:
        return 0
    return 1 + max(self.get_height(node.left), self.get_height(node.right))

def count_nodes(self, node):
    if node == self.NULL:
        return 0
    return 1 + self.count_nodes(node.left) + self.count_nodes(node.right)

# ===== SIMULASI =====
```

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```
# ===== SIMULASI =====

# WORST CASE (data terurut)
tree1 = RedBlackTree()
for i in range(1, 101):
    tree1.insert(i)

h1 = tree1.get_height(tree1.root)
n1 = tree1.count_nodes(tree1.root)

print("WORST CASE")
print("Node:", n1)
print("Height:", h1)
print("log2(n):", math.log2(n1))
print("2log2(n):", 2 * math.log2(n1))

# AVERAGE CASE (data acak)
tree2 = RedBlackTree()
data = list(range(1, 101))
random.shuffle(data)

for x in data:
```

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```
[ ] data = list(range(1, 101))
    random.shuffle(data)

    for x in data:
        tree2.insert(x)

    h2 = tree2.get_height(tree2.root)
    n2 = tree2.count_nodes(tree2.root)

    print("\nAVERAGE CASE")
    print("Node:", n2)
    print("Height:", h2)
    print("log2(n):", math.log2(n2))
```

... WORST CASE
Node: 100
Height: 11
log2(n): 6.643856189774724
2log2(n): 13.287712379549449

AVERAGE CASE
Node: 100
Height: 8
log2(n): 6.643856189774724

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